

VARUN MEDAPPA

CONTACT

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Skills

- Languages
 - Python
 - Bash
 - PyTorch
 - TensorFlow
 - SQL
- Platforms
 - AWS
 - Databricks
 - GCP

Accolades

- 2021 Best AI project award at a Vista-organized hackathon
- Patent for Multimodal Classification : <https://patents.justia.com/patent/12217493>

Education

Bachelor of Engineering, Industrial Engineering

R. V. College Of Engineering | 2011 - 2015

Master of Science, Operations Research

Columbia University | 2017 - 2019

PROFESSIONAL SUMMARY

Machine learning engineer and cross-functional leader with 9+ years in data science and ML, taking applied AI from an ambiguous problem to a working prototype to production. Founding member and hands-on leader of IAS's Multimodal ML team, building systems that detect misinformation, deepfakes and unsafe content across TikTok, YouTube, Meta and X at 3 trillion impressions a year. My work sits at the seam of ML engineering, research and policy: translating harm definitions into production classifiers, fine tuning and evaluating models, and turning proofs of concept into company-wide platforms, one of which cut labeling cost by 99.5%.

EXPERIENCE

Integral Ad Science | New York, NY | 2019 - Present

Staff Data Scientist - Multimodal Trust & Safety

- Ideated, prototyped and shipped ALP, an automated labeling platform (text, image and video embeddings, vector index, ANN search, VLM/LLM review of positive matches via OpenAI APIs). Took it from proof of concept to a company-wide capability that cut labeling cost and turnaround by 99.5%.
- Owned ALP's technical direction as a cross-team platform: prioritized features against a research roadmap and drove adoption across multiple teams.
- Took System One decision models (open-source alternatives to TypeSafe AI's Jev) from idea to production in one week: benchmarked variants, scaled scoring on AWS Auto Scaling Groups and pitched the CTO and leadership on putting it in front of our classifiers, cutting an estimated ~70% of costs on production data.
- Served as interim manager of the Innovation team, leading 8 FTE across Trust & Safety policy, ML research, ML engineering and DevOps, aligning priorities with product leadership.

Senior Data Scientist - Multimodal Trust & Safety

- Directed misinformation detection end to end: prototyped a teacher model using open-source LLMs with retrieval-augmented generation (RAG) to label training data, trained and productionized DistilBERT models extended for long token lengths on those labels to flag misleading narratives across 22M+ videos per day.
- Led adoption of PEFT/LoRA fine tuning for audio/video deepfake detection, an emerging-abuse workstream, cutting compute by ~80% and raising experiment throughput ~3x with no loss in detection performance.
- As a founding member of the Multimodal ML (Innovation/Media) team, built its multimodal video harm detection system (TensorFlow, ECS Fargate) covering 12 GARM unsafe-content and 37 contextual categories. It now processes 50 years of video per day and 3 trillion impressions per year across TikTok, YouTube, Meta and X, powering ~\$150M in annual revenue (~30% of company). Patented as US 12,217,493.

Data Scientist - Multimodal Trust & Safety

- Developed custom machine translation models (OpenNMT) for 42 language pairs, translating 75 trillion characters per year at 67% lower cost and enabling multilingual harm detection in production pipelines.
- Built a pseudo-labeling pipeline by fine tuning vision-language models (SigLIP, Vicuna), cutting model development cost by 90%+ across 12 unsafe categories, speeding coverage of new harm categories.
- Led a team generating synthetic Twitter data with LLMs (GPT-Neo), raising harmful-content classifier precision by 49% and reducing false positives.

Associate Data Scientist - Incrementality/Lift team

- Supported a conversion lift pipeline ingesting 8B impressions/day to measure ad campaign effects.

Capillary Technologies | Bangalore | 2016-2017

Business Analyst

Mu Sigma | Bangalore | 2015-2016

Data Scientist